

Basic Code Analysis: SQL Query Performance & Indexing Analysis

Duration : 30 Min.

Scenario

A database query on orders (customer_id, status) takes 12 seconds due to full table scans. Inspect 'query_plan.txt' and identify the missing index column.

Target File Location & Creation

File to Create/Update: student_workspace/solution.json

Task Objectives

Perform the following actions inside the student_workspace directory: - Create/update 'solution.json' inside 'student_workspace/' with fields 'table_name' ('orders') and 'index_column' ('customer_id').

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Open and inspect the scenario input files in student_workspace/.
3. Create/update the file student_workspace/solution.json and populate it with valid parameters and configurations.
4. Save your changes (Ctrl + S or Cmd + S).
5. Open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.
6. Verify your work by running python run.py in the terminal before submitting.

Validation

Once you have saved your files and verified your progress, return to the platform dashboard and click the **"Run Test" / "Verify"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	solution.json exists in student_workspace/	3 Marks

TC2	Target table ('orders') identified	4 Marks
TC3	Missing index column ('customer_id') identified	3 Marks

Total Score: 10 Marks

Important Notes

- This is an auto-evaluated question. Ensure all code edits are properly saved and the 'run.py' checks pass before submission.